

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Football Match Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a specialized data layer designed to track soccer match entities, team memberships, and user followings. It implements a relational schema focused on match lifecycle state transitions, scoring validation, and user-match associations, hosted within a structured database environment using UUIDs for entity identification.

1.2 Maturity Assessment

The project is currently in a **REFINEMENT** state. While the data model is structurally complete from a schema perspective, there are high-severity gaps regarding business goals and scope definition. The lack of a defined project objective and the uncertainty surrounding the authentication implementation for the User entity prevent an immediate transition to a ready state.

1.3 Technical Stack

- **Identifiers:** UUID (128-bit universally unique identifiers)
- **Temporal Standard:** UTC (Coordinated Universal Time)
- **Data Types:** DateTime, Enum, Boolean, Integer, String, URL

1.4 Architectural Constraints

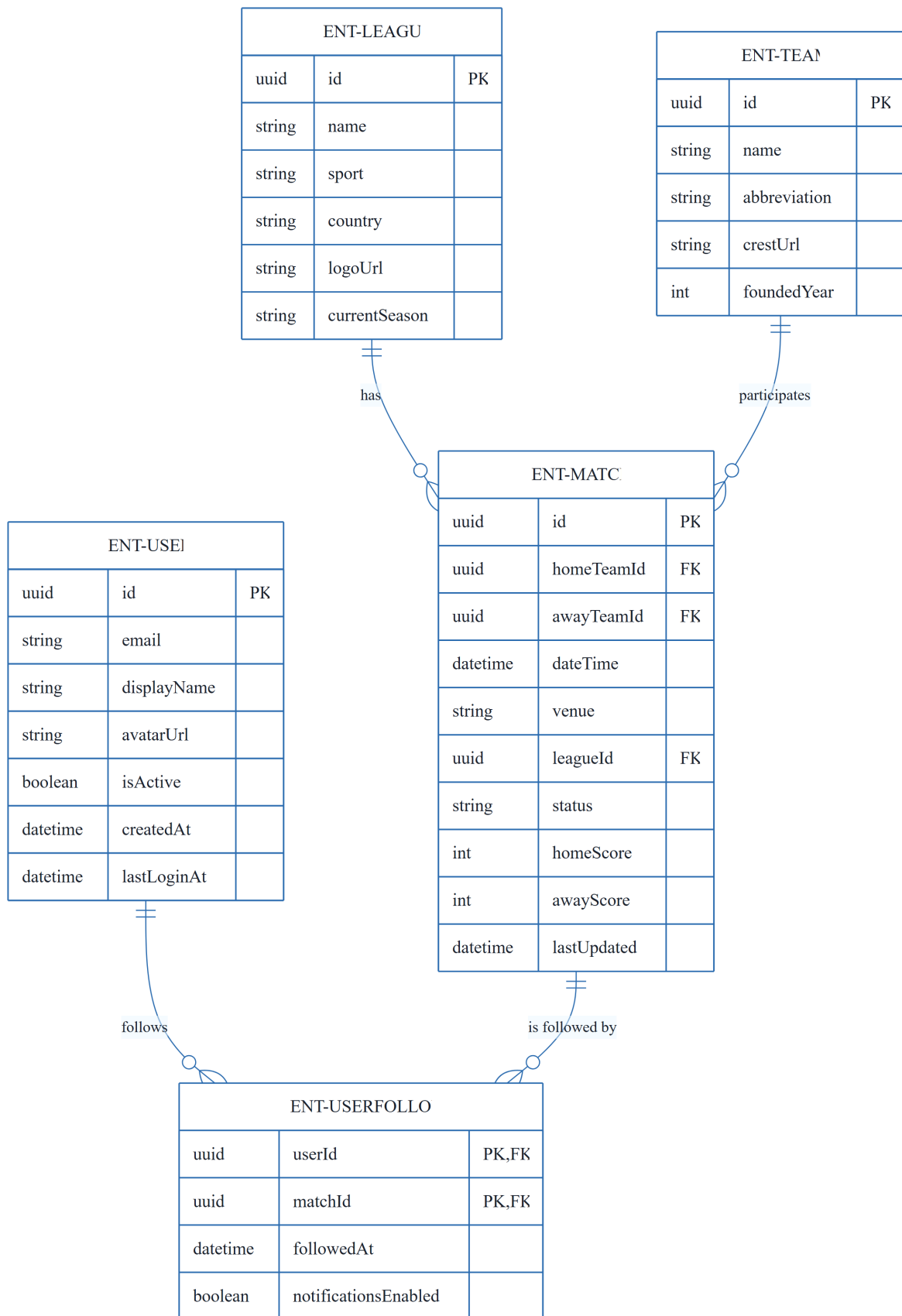
- **Temporal Logic:** Match `dateTime` must be in the future for 'upcoming' status.
- **State Integrity:** `homeScore` and `awayScore` must be non-null when status is 'finished'.
- **Value Constraints:** Scores must be strictly non-negative (min 0).
- **Entity Logic:** `homeTeamId` must not equal `awayTeamId`.
- **Uniqueness:** Team name must be globally unique; User email must be unique.
- **Relational Integrity:** `UserFollow` requires a composite unique constraint on (`userId`, `matchId`).
- **Domain Bounds:** Team `foundedYear` must be between 1800 and current year.
- **Field Limits:** Team abbreviation max 5 chars; Team/League name max 100 chars; Venue max 200 chars.

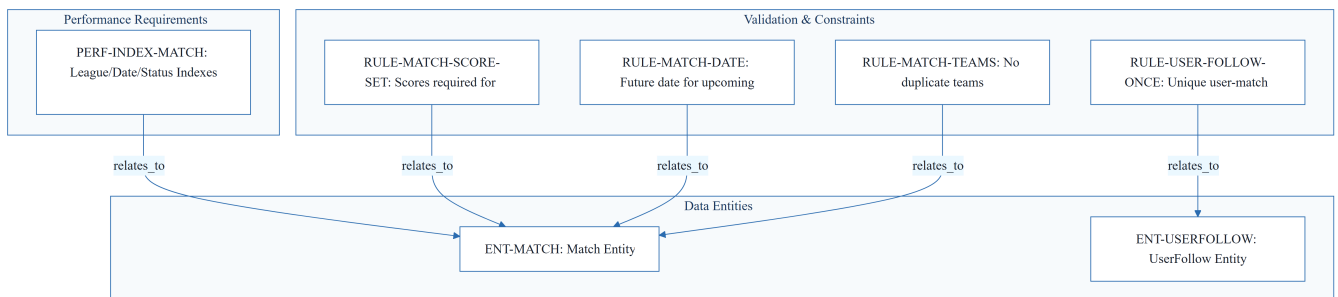
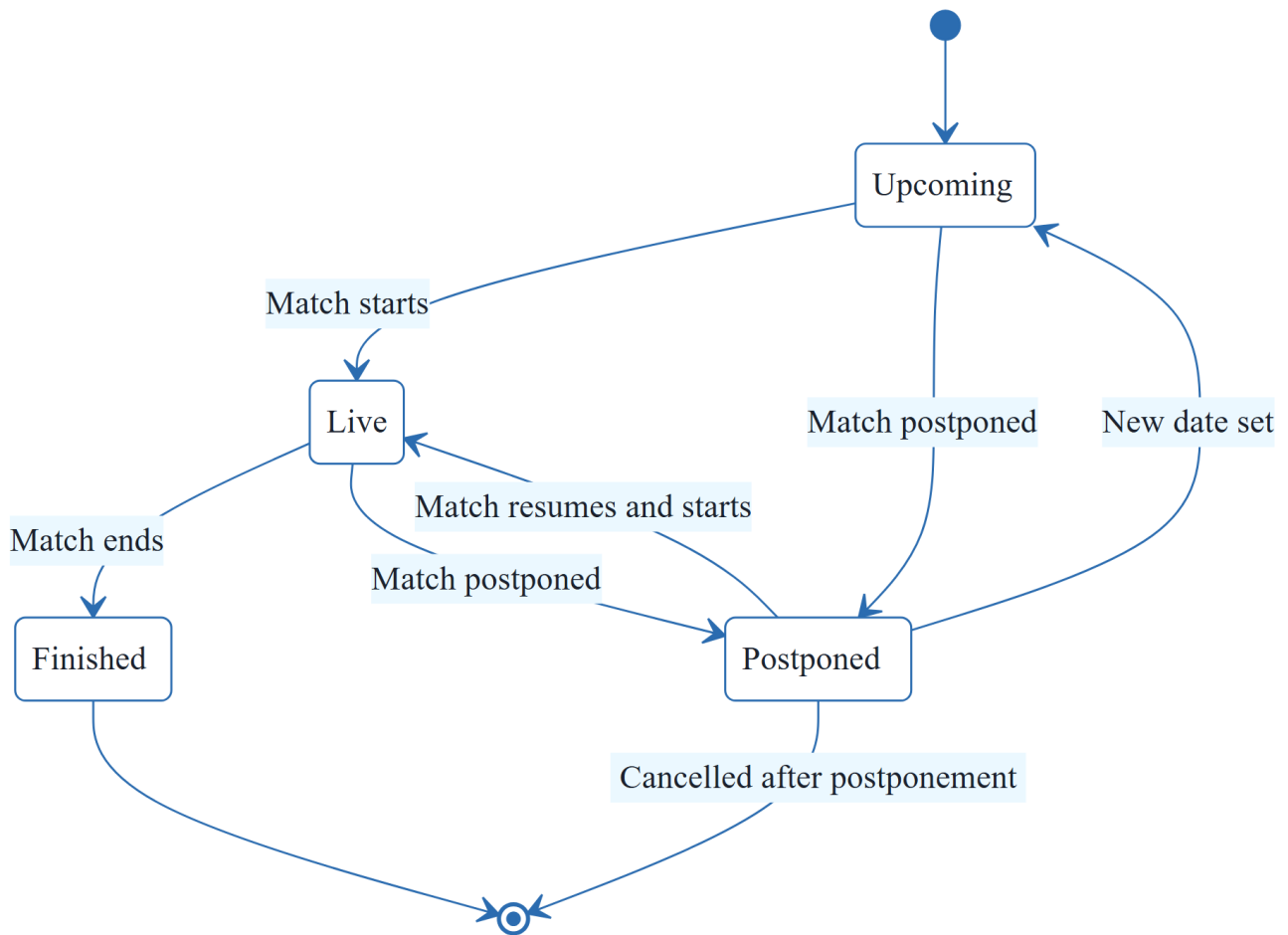
1.5 Critical Dependencies

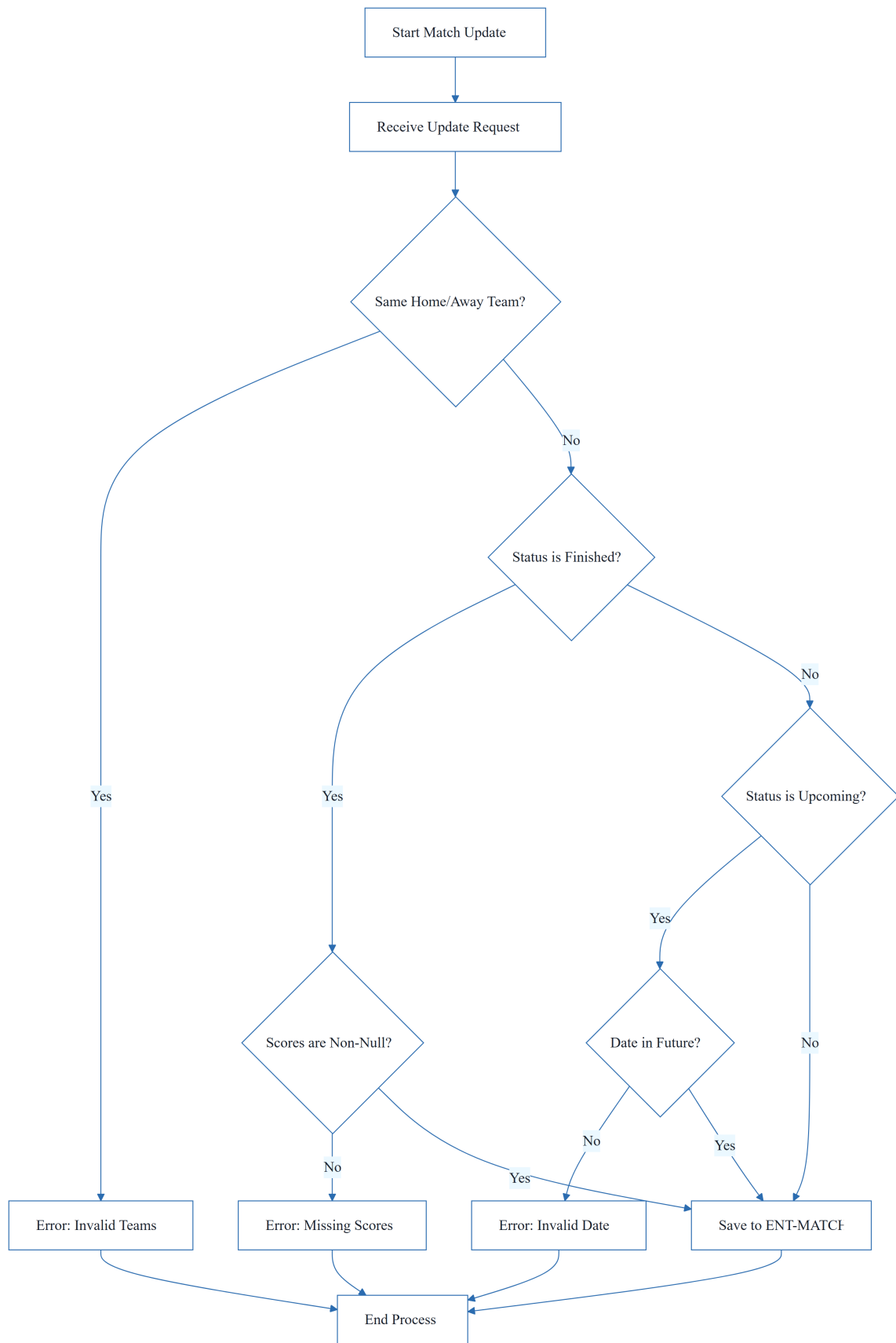
- **Referential Integrity:** ENT-MATCH depends on ENT-TEAM and ENT-LEAGUE entities.

- **Referential Integrity:** ENT-USERFOLLOW depends on ENT-USER and ENT-MATCH entities.
- **Composite Key Dependency:** UserFollow uniqueness bound to (userId, matchId).
- **Indexing Requirements:** (leagueId, dateTime) and (status) on ENT-MATCH for performance filtering.
- **Indexing Requirements:** (email) on ENT-USER for authentication lookups.

2. Architecture Workflows

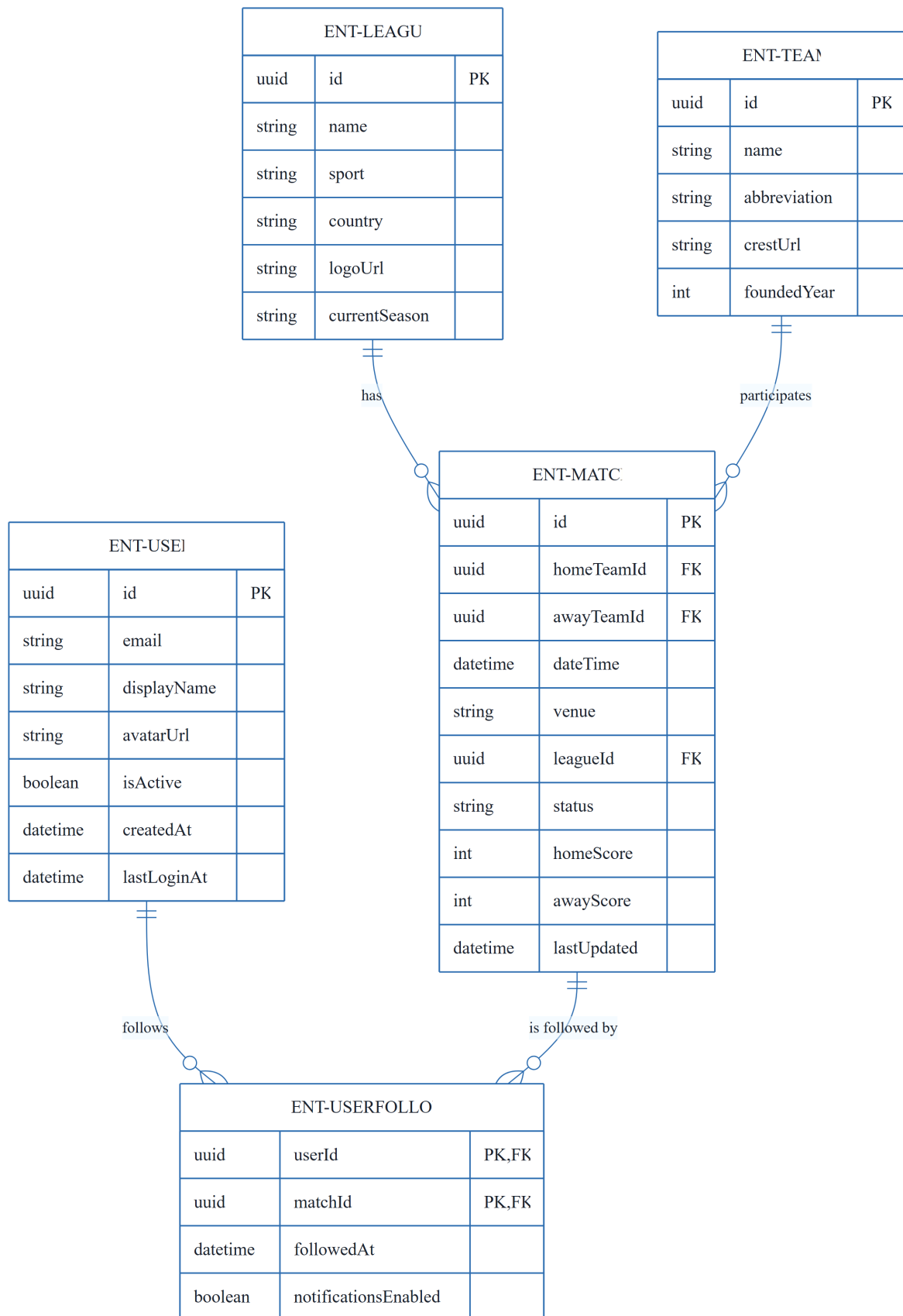




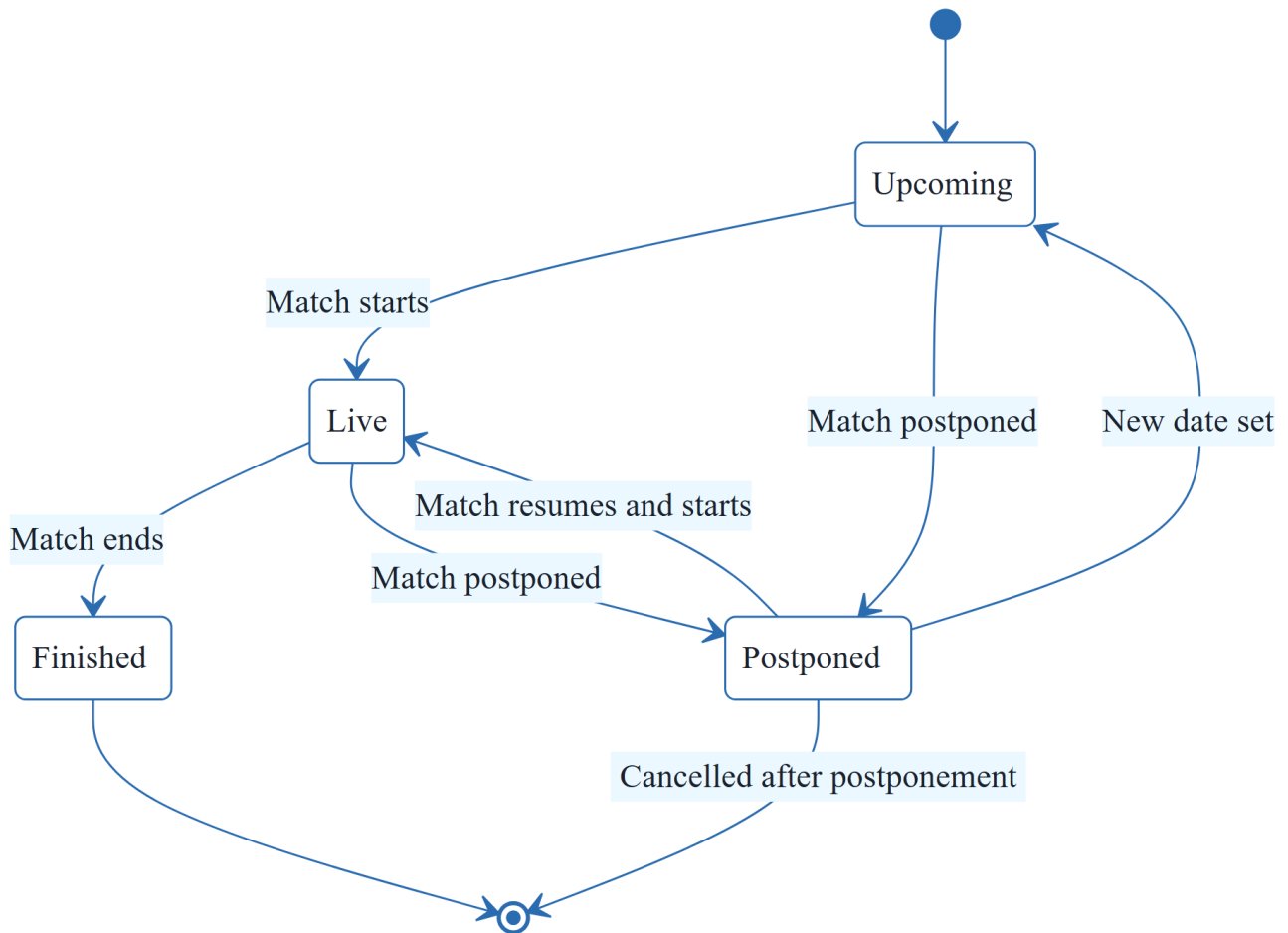


& Visual Diagrams

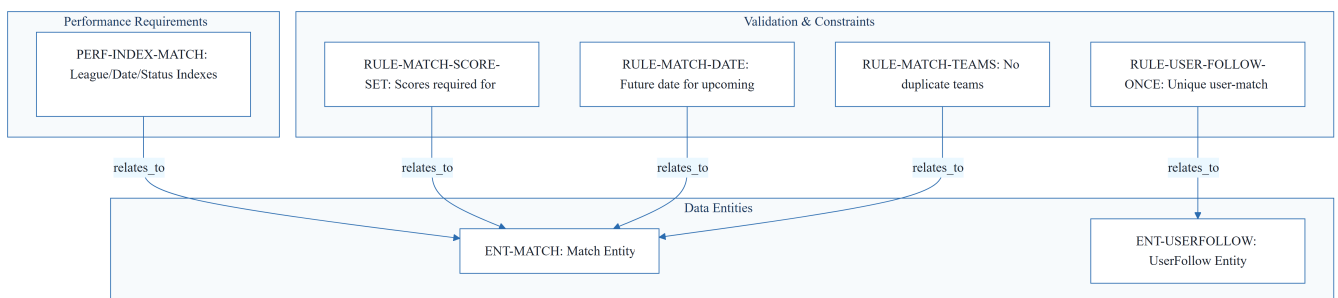
2.1 Data Model (ER Diagram)



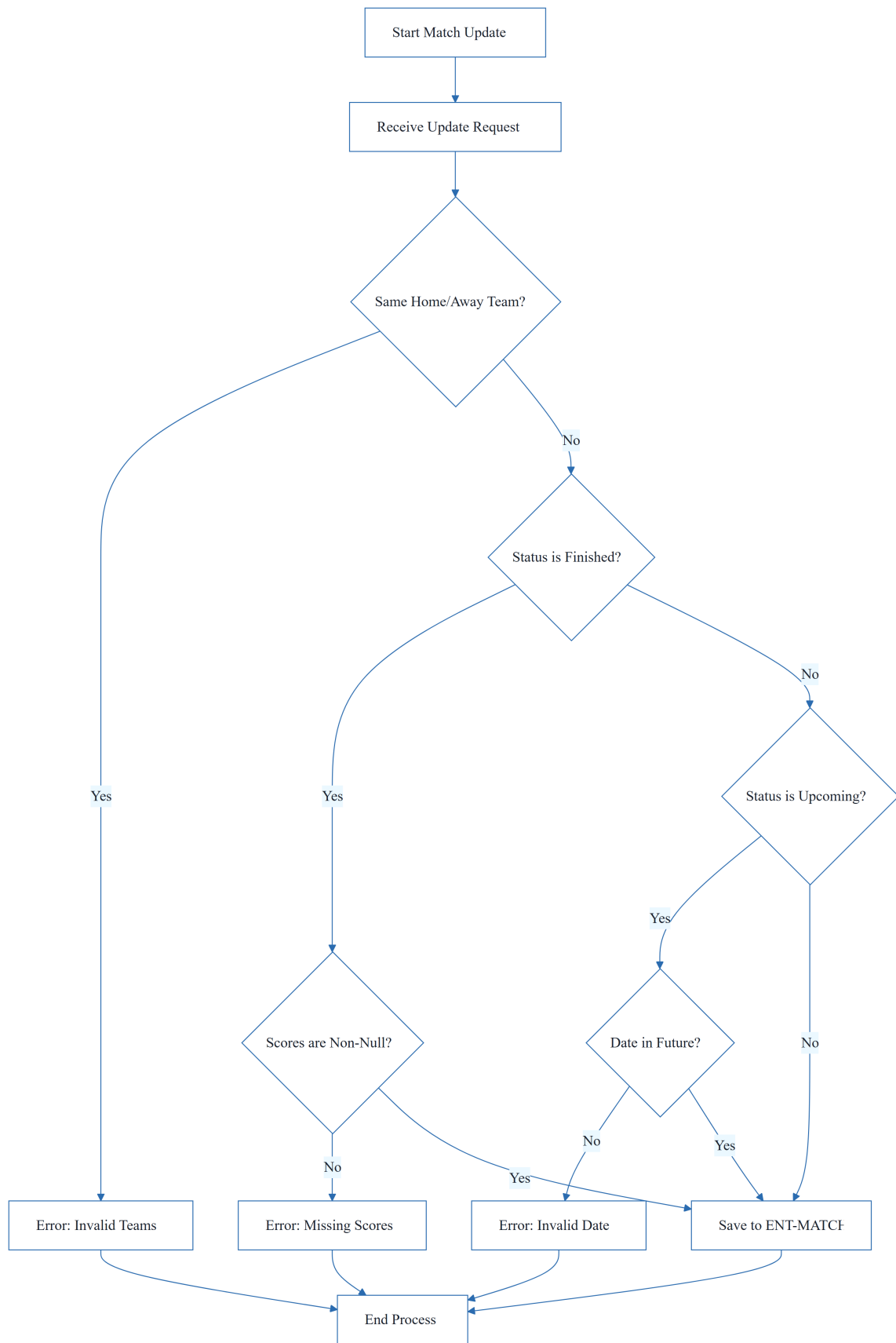
2.2 Match State Lifecycle



2.3 Requirements Traceability Matrix



2.4 Match Validation Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Requirement / Entity	Description	Source Section
ENT-MATCH	Entity: Match	Represents a football match with fields: id, homeTeamId, awayTeamId, dateTime, venue, leagueId, status, homeScore, awayScore, lastUpdated	Match
ENT-TEAM	Entity: Team	Represents a football team with fields: id, name, abbreviation, crestUrl, foundedYear	Team
ENT-LEAGUE	Entity: League	Represents a football league or tournament with fields: id, name, sport, country, logoUrl, currentSeason	League
ENT-USER	Entity: User	Represents an application user with fields: id, email, displayName, avatarUrl, isActive, createdAt, lastLoginAt	User
ENT-USERFOLLOW	Entity: UserFollow	Represents the relationship between a User and a Match they are following with fields: userId, matchId, followedAt, notificationsEnabled	UserFollow
RULE-MATCH-TEAMS	Functional Req	A match cannot have the same team as both home and away.	Validation Rules
RULE-MATCH-DATE	Functional Req	Match dateTime must be in the future for status 'upcoming'.	Validation Rules
RULE-MATCH-SCORE-SET	Functional Req	When status changes to 'finished', both homeScore and awayScore must be set (non-null).	Validation Rules
RULE-SCORE-POSITIVE	Constraint	Scores cannot be negative.	Validation Rules
RULE-TEAM-UNIQUE	Functional Req	Team name must be unique across all teams.	Validation Rules
RULE-USER-EMAIL-UNIQUE	Functional Req	Email must be unique.	Validation Rules

ID	Requirement / Entity	Description	Source Section
RULE-USER-FOLLOW-ONCE	Constraint	A user can only follow a match once (Composite unique constraint on userId, matchId).	UserFollow
PERF-INDEX-MATCH	Non-Functional Req	Indexes on (leagueId, dateTime) and (status) for performance filtering.	Indexes (for performance)

3.2 Security Rules

- **Identity Integrity:** User email must be unique (RULE-USER-EMAIL-UNIQUE).
- **Credential Complexity:** Passwords (if implemented) must meet complexity requirements (min 8 characters).
- **Access Control:** User account status is tracked via isActive boolean.

3.3 Data Models

ENT-MATCH

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique (PK)
homeTeamId	UUID	Reference to Team	Required, exists (FK)
awayTeamId	UUID	Reference to Team	Required, exists, != homeTeamId (FK)
dateTime	DateTime	Match date and time (UTC)	Required, future if 'upcoming'
venue	String	Stadium or location name	Optional, max 200
leagueId	UUID	Reference to League	Required, exists (FK)
status	Enum	upcoming, live, finished, postponed, cancelled	Required, default: upcoming
homeScore	Integer	Goals scored by home team	Optional, min 0, default null
awayScore	Integer	Goals scored by away team	Optional, min 0, default null
lastUpdated	DateTime	Timestamp of last update	Auto-set on update

ENT-TEAM

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique (PK)

Field	Type	Description	Validation
name	String	Team name	Required, unique, max 100
abbreviation	String	Team abbreviation	Optional, unique per league, max 5
crestUrl	URL	URL to team crest/logo	Optional, valid URL
foundedYear	Integer	Year the club was founded	Optional, 1800 <= year <= current

ENT-LEAGUE

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique (PK)
name	String	League name	Required, max 100
sport	String	Sport type	Required, default: "football"
country	String	Country where league is based	Optional, max 100
logoUrl	URL	URL to league logo	Optional, valid URL
currentSeason	String	Current season identifier	Optional (e.g., "2023/24")

ENT-USER

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique (PK)
email	String	User email address	Required, unique, valid email
displayName	String	Display name for the user	Required, max 50
avatarUrl	URL	URL to user avatar/image	Optional, valid URL
isActive	Boolean	Whether the user account is active	Required, default: true
createdAt	DateTime	Account creation timestamp	Auto-set on create
lastLoginAt	DateTime	Last login timestamp	Updated on login

ENT-USERFOLLOW

Field	Type	Description	Validation
userId	UUID	Reference to User	Required, exists (PK, FK)

Field	Type	Description	Validation
matchId	UUID	Reference to Match	Required, exists (PK, FK)
followedAt	DateTime	Timestamp of following	Required, auto-set on create
notificationsEnabled	Boolean	Notification preference	Optional, default: false

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	The document is purely technical/data-oriented. A section explaining the business purpose of the Match Manager is needed.
Scope & Out-of-Scope	MEDIUM	Define whether this data model covers only the match engine or also user profiles, payments, etc.
Open Questions & Uncertainties	LOW	The document mentions 'if authentication is implemented', which should be explicitly listed as an open question.

4.2 Remediation & Workflow

- Business Alignment:** Conduct a stakeholder workshop to define the "Goals & Objectives" and "Scope".
- Auth Decision:** Finalize the decision on whether the ENT-USER entity will be integrated with a full authentication system or remain a simple profile store.
- Validation Refinement:** Clarify if ENT-TEAM name uniqueness is global or scoped per league.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Constraints	TECHNICAL_STACK	RULE-USER-FOLLOW-ON CE	Restrictive database rules, specifically the composite unique requirement preventing duplicate entries for the same pair of associated identifiers.

Term	Category	Context Anchor	Project Definition
Cryptographic Hashing	TECHNICAL_STACK	RULE-USER-EMAIL-UNIQUE	The implicit security mechanism required to protect stored credentials mentioned in the complexity requirements for account access.
DateTime	TECHNICAL_STACK	ENT-MATCH	A temporal data type used for scheduling events and recording system timestamps.
MUN	BUSINESS_DOMAIN	ENT-TEAM	An example of a short-form identification string for a specific athletic club.
Match Validation	BUSINESS_DOMAIN	RULE-MATCH-TEAMS	The set of business rules ensuring logical consistency for sporting events, such as forbidding a team from playing against itself and requiring final scores upon completion.
Relationships	TECHNICAL_STACK	ENT-MATCH	The structural links between different data entities, including one-to-many and many-to-many associations.
Team Validation	BUSINESS_DOMAIN	RULE-TEAM-UNIQUE	Operational checks verifying that no two athletic organizations share the same identity name.
UTC	TECHNICAL_STACK	ENT-MATCH	The standardized global time reference used to avoid ambiguity across different geographical time zones.
UUID	TECHNICAL_STACK	ENT-MATCH	A 128-bit universally unique identifier used as the primary key for all major system entities.
User Validation	BUSINESS_DOMAIN	RULE-USER-EMAIL-UNIQUE	The integrity checks performed on account data, focusing on the uniqueness of contact addresses and credential strength.
UserFollow	BUSINESS_DOMAIN	ENT-USERFOLLOW	A junction entity tracking the subscription interest of a profile toward a specific sporting event.

Term	Category	Context Anchor	Project Definition
UserFollow Validation	BUSINESS_DOMAIN	RULE-USER-FOLLOW-ON CE	The specific logic ensuring a single account cannot subscribe to the same event more than once.